

Infosys Springboard Internship — Individual Contribution Documentation

Milestone 4: Week 7 & 8 — Testing, Deployment & Documentation

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1. Executive Summary & Overview

During Milestone 4 (Week 7 & 8), the core objective focused on prediction model validation audit, user interface responsiveness optimization, cloud containerization via Docker, Vercel SPA routing fallback, comprehensive system documentation, and complete end-to-end platform demonstration across all 4 system roles.

2. Key Modules & Individual Contribution

As an individual contributor, I designed, implemented, and verified the following technical modules:

- **Model Accuracy & System Validation Audit:** Verified Gradient Boosting classifier telemetry (97.5% Accuracy, 0.9867 F1-Score, 0.9960 ROC-AUC) under stress testing and edge-case patient parameters.
- **UI Responsiveness & Build Optimization:** Optimized Vite v6 compilation, sub-100ms dashboard re-renders, and full Tailwind CSS v4 white & crimson red medical design system compliance across 25+ views.
- **Cloud Containerization & Deployment Setup:** Engineered multi-stage Dockerfile, docker-compose configuration, and vercel.json SPA routing fallback for seamless cloud deployment.
- **Comprehensive Final Project Documentation:** Authored technical architecture schemas, REST API endpoint specifications, RBAC security matrices, and master milestone reports.
- **End-to-End Multi-Role Platform Demonstration:** Executed and verified end-to-end clinical workflows across Doctor, Hospital Admin, Researcher, and System Admin role portals.

■ Milestone 4 Outcomes Verification Matrix

Required Task / Deliverable	Target Outcome	Implementation Component	Status
Validate prediction accuracy & analytics	97.5% model precision, boundary tests & audit logs	/system-admin/models	COMPLETED
Optimize healthcare workflows & UI	Sub-100ms re-renders, Vite build & responsive layouts	Frontend / Tailwind v4	COMPLETED
Deploy platform using Docker & Cloud	Multi-stage Dockerfile, Compose & Vercel SPA config	Docker / Vercel	COMPLETED
Prepare final documentation & presentation	Complete architecture schema, API docs & reports	System Docs	COMPLETED
Demonstrate complete platform	Verified end-to-end multi-role clinical workflows	All 4 Workspaces	COMPLETED